Options :

6406534314786. ✖ [64, 784]

6406534314787. ✖ [64, 100]

6406534314788. ✔ [64, 884]

6406534314789. ✖ [128, 784]

Speech Technology

Section Id :	64065391717
Section Number :	9
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	2
Number of Questions to be attempted :	2
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653202035
Question Shuffling Allowed :	No

Question Number : 168 Question Id : 6406531279878 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534314790. ✔ YES

6406534314791. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653202036
Question Shuffling Allowed :	No

Question Number : 169 Question Id : 6406531279879 Question Type : MCQ

Correct Marks : 25

Question Label : Multiple Choice Question

1. (1 point) Short Time Fourier Transform (STFT) is required to analyze speech because speech is

   (a) quasi-stationary

   (b) stationary

   (c) non-stationary

2. (1 point) Voiced sounds result in speech signals.

   (a) periodic

   (b) aperiodic

3. (1 point) For perfect reconstruction of a bandlimited signal, the sampling frequency must be

   (a) equal to twice the maximum frequency

   (b) greater than or equal to twice the maximum frequency

   (c) lesser than or equal to twice the maximum frequency

   (d) none of the above

4. (1 point) Which of the following best describes the role of the Mel-Frequency Cepstral Coefficients (MFCC) in speech processing?

   (a) They capture pitch information directly.

   (b) They represent the short-term power spectrum on a perceptual scale.

   (c) They measure phase variations.

   (d) They enhance the high-frequency components.

5. (3 points) Comment on the linearity and time-invariance of the following systems?

   (a) $y[n] = 2x[n] + 3$

   (b) $y[n] = x[n]^2$

   (c) $y[n] = x[2n]$

6. (3 points) **KMeans clustering** You are given the following 1-dimensional data points:

$$X = \{2,\ 4,\ 5,\ 8,\ 9\}$$

   (a) Initialize 2 cluster centroids: $c_1 = 4$ and $c_2 = 8$.

   (b) Assign each data point to the nearest cluster.

   (c) Compute the new centroids after assignment.

   (d) Write the new clusters and their updated centroids.

7. (5 points) In speech coding, the **Linde-Buzo-Gray (LBG)** algorithm is used to design a vector quantizer for compressing feature vectors.

   Given the following 1-dimensional training data points:

$$X = \{1,\ 2,\ 3,\ 8,\ 9,\ 10\}$$

   (a) Perform **one iteration** of the LBG algorithm starting with a single centroid at the global mean.

   (b) Apply the **splitting step** to obtain two centroids.

   (c) Reassign the training data to the nearest centroid and compute the new centroids.

   Hint: Use a small perturbation factor $\epsilon = 0.1$ for splitting:

$$\text{Split:} \quad c_1 = \mu(1+\epsilon), \quad c_2 = \mu(1-\epsilon)$$

8. (5 points) In speech recognition, Gaussian Mixture Models (GMMs) are used to model the probability distribution of feature vectors such as MFCCs.

   Suppose you have a 1-dimensional MFCC feature $x = 2.0$. The feature is modeled using a **2-component GMM** with the following parameters:

   - Component 1: $\mu_1 = 1.0,\ \sigma_1^2 = 1.0,\ w_1 = 0.6$
   - Component 2: $\mu_2 = 3.0,\ \sigma_2^2 = 1.0,\ w_2 = 0.4$

   (a) Compute the probability density $p(x)$ of x under this GMM.

   (b) Compute the posterior probability that the feature x was generated by **Component 1**.

   Hint: The PDF of a 1D Gaussian is

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$$

9. (5 points) We discussed two approaches for decoding in Hidden Markov Models (HMMs): the **greedy** approach and the **Viterbi** algorithm.

Consider a part-of-speech (POS) tagging problem modeled as a first-order HMM. Given the observed word sequence: *"Time flies fast"* and that the first word "Time" is tagged as a **Noun**, determine the most probable tag sequence for the next two words using:

(a) the greedy approach

(b) the Viterbi algorithm

Hint: In the Viterbi approach, keep track of the best path probabilities for each possible tag at each position.

Assume the following transition and emission probabilities:

Transition	To Noun	To Verb	To Adjective
From Noun	0.3	0.5	0.2
From Verb	0.4	0.4	0.2
From Adjective	0.2	0.3	0.5

Word	P(Word—Noun)	P(Word—Verb)	P(Word—Adjective)
flies	0.1	0.8	0.1
fast	0.2	0.3	0.5

Options :

6406534314792. ✔ I have written answers on the answer sheets

6406534314793. ✖ Not applicable

Maths2

Section Id :	64065391718
Section Number :	10
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes

Maximum Instruction Time : 0
Sub-Section Number : 1
Sub-Section Id : 640653202037
Question Shuffling Allowed : No

Question Number : 170 Question Id : 6406531279880 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "FOUNDATION LEVEL : MATHEMATICS FOR DATA SCIENCE II (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406534314794. ✔ YES
6406534314795. ✖ NO

Sub-Section Number : 2
Sub-Section Id : 640653202038
Question Shuffling Allowed : Yes

Question Number : 171 Question Id : 6406531279881 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Consider the following statements about the function $T : \mathbb{R}^2 \to \mathbb{R}$ defined by $T(x, y) = 2x + y + 1$, for all $(x, y) \in \mathbb{R}^2$.

Statement P: T is a linear transformation.

Statement Q: T is a one-to-one (injective) map.

Choose the correct option from the following.

Options :
6406534314796. ✖ Both P and Q are true.
6406534314797. ✖ P is true and Q is false.
6406534314798. ✖ P is false and Q is true.
6406534314799. ✔ Both P and Q are false.

Sub-Section Number : 3
Sub-Section Id : 640653202039
Question Shuffling Allowed : Yes

Question Number : 172 Question Id : 6406531279882 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is a basis for $V = \{A \in M_{2\times 2}(\mathbb{R}) \mid A^T = A\}$, the vector space of 2×2 real symmetric matrices?

Options :

6406534314800. ✖ $\left\{\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}, \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\right\}$

6406534314801. ✖ $\left\{\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\right\}$

6406534314802. ✔ $\left\{\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}, \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}\right\}$

6406534314803. ✖ $\left\{\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}\right\}$

Sub-Section Number :	4
Sub-Section Id :	640653202040
Question Shuffling Allowed :	Yes

Question Number : 173 Question Id : 6406531279883 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

Choose the matrix which satisfies the equation $det(A - cI) = c^2 - 5c + 6$.

Options :

6406534314804. ✖ $A = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix}$

6406534314805. ✖ $A = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix}$

6406534314806. ✔ $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$

6406534314807. ✖ $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

Sub-Section Number : 5
Sub-Section Id : 640653202041
Question Shuffling Allowed : Yes

Question Number : 174 Question Id : 6406531279884 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Given a vector space V, which of the following statements are equivalent to the statement that a set $\mathcal{B} \subseteq V$ is a basis?

Options :

6406534314808. ✔ $\mathcal{B}$ is a maximal linearly independent set of V.

6406534314809. ✖ $\mathcal{B}$ is a minimal linearly independent set of V.

6406534314810. ✖ $\mathcal{B}$ is a maximal spanning set of V.

6406534314811. ✔ $\mathcal{B}$ is a minimal spanning set of V.

Sub-Section Number : 6
Sub-Section Id : 640653202042
Question Shuffling Allowed : Yes

Question Number : 175 Question Id : 6406531279885 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0
Question Label : Multiple Select Question

Let A be a 3×3 matrix such that $\det(A) = -3$. For which of the options below is the matrix B, obtained from the matrix A by performing the operations in the option, such that $\det(B) = 12$?

Options :

6406534314812. ✔ B is obtained by swapping the first two rows and then multiplying the third row by 4.

6406534314813. ✖ B is obtained by swapping the first and third row and then multiplying the second row by -4.

6406534314814. ✔ B is obtained by multiplying the second row by -4.

6406534314815. ✔ B is obtained by multiplying the first row by 2 and the third row by -2.

Question Number : 176 Question Id : 6406531279886 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0
Question Label : Multiple Select Question

Each of the options below contains a set V along with an addition operation $+ : V \times V \to V$ and a scalar multiplication operation $\cdot : \mathbb{R} \times V \to V$. Choose the options for which there is an additive identity in V with respect to $+$.

Options :

6406534314816. ✔ $V = \{x \in \mathbb{R} \mid x > 0\}$, $x + y = xy$, $c \cdot x = x^c$, for all $x, y \in V$ and $c \in \mathbb{R}$.

6406534314817. ✔ $V = \{(x, 1) \mid x \in \mathbb{R}\}$, $(x, 1) + (y, 1) = (x + y, 1)$, $c \cdot (x, 1) = (cx, 1)$, for all $x, y, c \in \mathbb{R}$.

6406534314818. ✖ For an $m \times n$ matrix A and a non-zero vector b in its column space, $V = \{x \in \mathbb{R}^n \mid Ax = b\}$, and $+, \cdot$ denote the usual addition and scalar multiplication of vectors in $\mathbb{R}^n$, respectively.

6406534314819. ✔ V is the set of skew-symmetric matrices of order n, i.e., $V = \{A \in M_{n \times n}(\mathbb{R}) \mid A^T = -A\}$, and $+, \cdot$ denote the usual addition and scalar multiplication of $n \times n$ matrices, respectively.

Sub-Section Number : 7
Sub-Section Id : 640653202043

Question Shuffling Allowed : No

Question Id : 6406531279887 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (177 to 178)

Question Label : Comprehension

Let $A = \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix} \in M_{1\times 4}(\mathbb{R})$ be a non-zero matrix and M denote the reduced row echelon form of $A^T A$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 177 Question Id : 6406531279888 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Find the rank of *A*.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 178 Question Id : 6406531279889 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Find the number of zero rows of *M*.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Sub-Section Number : 8

Sub-Section Id : 640653202044

Question Shuffling Allowed : No

Question Id : 6406531279890 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (179 to 180)

Question Label : Comprehension

Consider the vectors $(k, 2, -1), (0, k-1, 4)$ and $(0, 3, k-2)$ in $\mathbb{R}^3$, for some $k \in \mathbb{R}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 179 Question Id : 6406531279891 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Find the number of values of k for which the given three vectors are linearly dependent in $\mathbb{R}^3$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 180 Question Id : 6406531279892 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Find the largest value of k for which the given three vectors are linearly dependent in $\mathbb{R}^3$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number :	9
Sub-Section Id :	640653202045
Question Shuffling Allowed :	No

Question Id : 6406531279893 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (181 to 183)

Question Label : Comprehension

Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & -1 & -1 \\ -2 & 1 & 0 & k \\ 4 & -1 & k & 0 \end{bmatrix},$$

for some $k \in \mathbb{R}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 181 Question Id : 6406531279894 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Find the maximum possible rank of the matrix *A*.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 182 Question Id : 6406531279895 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Find the maximum possible nullity of the matrix *A*.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 183 Question Id : 6406531279896 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Find the value of *k* for which rank of the matrix *A* is equal to 2.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
-2

Sub-Section Number :	10
Sub-Section Id :	640653202046
Question Shuffling Allowed :	No

Question Id : 6406531279897 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (184 to 186)

Question Label : Comprehension

Shakti bought 2 pencils and 3 pens from a shop and paid Rs. a to the shopkeeper.
Surya bought b pencils and 6 pens and paid Rs. 10 to the shopkeeper.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 184 Question Id : 6406531279898 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Let x_1 be the price of 1 pencil and x_2 be the price of 1 pen.
Choose the correct option.

Options :

The matrix representation to find x_1 and x_2 is:

$$\begin{bmatrix} 2 & b \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a \\ 10 \end{bmatrix}$$

6406534314827. ✖

The matrix representation to find x_1 and x_2 is:

$$\begin{bmatrix} 2 & 3 \\ b & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a \\ 10 \end{bmatrix}$$

6406534314828. ✔

The system of equations

$$2x_1 + 6x_2 = a$$
$$bx_1 + 3x_2 = 10$$

6406534314829. ✖ represents the given problem.

The system of equations

$$2x_1 + bx_2 = a$$
$$3x_1 + 6x_2 = 10$$

6406534314830. ✖ represents the given problem.

Question Number : 185 Question Id : 6406531279899 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Suppose Shakti tries to find the price of 1 pencil and 1 pen and he hopes to have unique values for the prices.
Then b should **not** be equal to

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
4

Question Number : 186 Question Id : 6406531279900 Question Type : SA
Correct Marks : 4
Question Label : Short Answer Question

Suppose Shakti chooses some values for a and b such that multiple sets of values could represent the prices of 1 pencil and 1 pen.
What should the value of a be?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
5

Statistics2

Section Id :	64065391719
Section Number :	11
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No

Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653202047
Question Shuffling Allowed :	No

Question Number : 187 Question Id : 6406531279901 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "FOUNDATION LEVEL : STATISTICS FOR DATA SCIENCE II (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534314833. ✔ YES

6406534314834. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653202048
Question Shuffling Allowed :	No

Question Number : 188 Question Id : 6406531279902 Question Type : MCQ
Correct Marks : 0

Question Label : Multiple Choice Question

Discrete random variables:

Distribution	PMF ($f_X(k)$)	CDF ($F_X(x)$)	$E[X]$	$\mathrm{Var}(X)$
Uniform(A) $A=\{a,a+1,\ldots,b\}$	$\frac{1}{n}, \quad x=k$ $n=b-a+1$ $k=a,a+1,\ldots,b$	$\begin{cases} 0 & x<0 \\ \frac{k-a+1}{n} & k\le x<k+1 \\ & k=a,a+1,\ldots,b-1,b \\ 1 & x\ge n \end{cases}$	$\frac{a+b}{2}$	$\frac{n^2-1}{12}$
Bernoulli(p)	$\begin{cases} p & x=1 \\ 1-p & x=0 \end{cases}$	$\begin{cases} 0 & x<0 \\ 1-p & 0\le x<1 \\ 1 & x\ge 1 \end{cases}$	p	$p(1-p)$
Binomial(n,p)	${}^nC_k p^k(1-p)^{n-k}$, $k=0,1,\ldots,n$	$\begin{cases} 0 & x<0 \\ \sum_{i=0}^{k} {}^nC_i p^i(1-p)^{n-i} & k\le x<k+1 \\ & k=0,1,\ldots,n \\ 1 & x\ge n \end{cases}$	np	$np(1-p)$
Geometric(p)	$(1-p)^{k-1}p$, $k=1,\ldots,\infty$	$\begin{cases} 0 & x<0 \\ 1-(1-p)^k & k\le x<k+1 \\ & k=1,\ldots,\infty \end{cases}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$\frac{e^{-\lambda}\lambda^k}{k!}$, $k=0,1,\ldots,\infty$	$\begin{cases} 0 & x<0 \\ e^{-\lambda}\sum_{i=0}^{k}\frac{\lambda^i}{i!} & k\le x<k+1 \\ & k=0,1,\ldots,\infty \end{cases}$	λ	λ

Continuous random variables:

Distribution	PDF ($f_X(k)$)	CDF ($F_X(x)$)	$E[X]$	$\mathrm{Var}(X)$
Uniform$[a,b]$	$\frac{1}{b-a}$, $a \le x \le b$	$\begin{cases} 0 & x \le a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x \ge b \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exp(λ)	$\lambda e^{-\lambda x}$, $x > 0$	$\begin{cases} 0 & x \le 0 \\ 1-e^{-\lambda x} & x > 0 \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$, $-\infty < x < \infty$	No closed form	μ	σ^2
Gamma(α, β)	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, $x > 0$		$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$ $0 < x < 1$		$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

1. **Markov's inequality:** Let X be a discrete random variable taking non-negative values with a finite mean μ. Then,
$$P(X \ge c) \le \frac{\mu}{c}$$

2. **Chebyshev's inequality:** Let X be a discrete random variable with a finite mean μ and a finite variance σ^2. Then,
$$P(|X - \mu| \ge k\sigma) \le \frac{1}{k^2}$$

Useful data:

1. Use the following values if required:
$e^{-3} = 0.0498$, $e^{-4} = 0.0183$, $e^{-2} = 0.1353$, $e^{-1.2} = 0.3011$,
$e^{-0.83} = 0.4360$, $e^{-0.2} = 0.8187$,

2. $\int x^n dx = \frac{x^{n+1}}{n+1}$.

Options :

6406534314835. ✔ Useful Data has been mentioned above.

6406534314836. ✖ This data attachment is just for a reference & not for an evaluation.

Sub-Section Number : 3

Sub-Section Id : 640653202049

Question Shuffling Allowed : Yes

Question Number : 189 Question Id : 6406531279903 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Let X_1, X_2 and X_3 be three independent and identically distributed Poisson random variables with mean $\lambda = 3$. Find the probability that exactly one of the X_i is equal to 0, and other two variables are greater than 1.

Options :

6406534314837. ✔ $3e^{-3}(1-4e^{-3})^2$

6406534314838. ✖ $6e^{-3}(1-4e^{-3})$

6406534314839. ✖ $e^{-3}(1-4e^{-3})^2$

6406534314840. ✖ $6e^{-6}(1-4e^{-3})^2$

Question Number : 190 Question Id : 6406531279904 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Let X be the number of items produced by a machine on a given day with $E[X] = 7$ and $E[X^2] = 55$. Using Chebyshev's inequality, find a lower bound on the probability that a machine produces between 4 and 10 items on a given day.

Options :

6406534314841. ✖ $\frac{2}{3}$

6406534314842. ✔ $\frac{1}{3}$

6406534314843. ✖ $\frac{1}{2}$

6406534314844. ✖ $\frac{3}{4}$

Sub-Section Number :	4
Sub-Section Id :	640653202050
Question Shuffling Allowed :	Yes

Question Number : 191 Question Id : 6406531279905 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The time (in minutes) a customer spends inside a service center follows an exponential distribution with an average of 10 minutes. What is the probability that a customer has to wait at least 12 more minutes in the center given that he has already spent 10 minutes in the center? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.27 to 0.33

Question Number : 192 Question Id : 6406531279906 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

The joint PMF of two discrete random variables X and Y is given by:

$$f_{XY}(x,y) = \begin{cases} \frac{1}{14}(2x+3y+1), & x,y \in \{0,1\} \\ 0, & \text{otherwise.} \end{cases}$$

What is the value of $P(0 < X \leq 1 \mid Y > 0)$? Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.4 to 0.8

Sub-Section Number : 5

Sub-Section Id : 640653202051

Question Shuffling Allowed : No

Question Id : 6406531279907 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (193 to 194)

Question Label : Comprehension

Let X be a continuous random variable with the following PDF:

$$f_X(x) = \begin{cases} \frac{x}{2}, & 0 \le x \le 2 \\ 0, & \text{otherwise.} \end{cases}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 193 Question Id : 6406531279908 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Calculate the value of Var*(X)*.

Options :

6406534314847. ✖ 2

6406534314848. ✖ $\frac{4}{3}$

6406534314849. ✔ $\frac{2}{9}$

6406534314850. ✖ $\frac{2}{3}$

Question Number : 194 Question Id : 6406531279909 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Define $Y = 72X + 5$, then find the value of $\text{Var}(Y)$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1152

Sub-Section Id :	640653202052
Question Shuffling Allowed :	No

Question Id : 6406531279910 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (195 to 196)

Question Label : Comprehension

A hospital tracks the number of patients arriving in the Emergency Room (X) per hour and the number of doctors available (Y) per hour. Suppose $X \sim \text{Poisson}(3)$ and Y is independent of X, with the following PMF:

Y	1	2	3
$P(Y = y)$	0.3	0.5	0.2

Let $Z = X - Y$. Assume that the hospital cannot treat more patients than the number of available doctors.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 195 Question Id : 6406531279911 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following expressions correctly represents the event that an arriving patient cannot be treated?

Options :

6406534314852. ✖ $X = 0$

6406534314853. ✔ $X > Y$

6406534314854. ✖ $Y > X$

6406534314855. ✖ $Z = 0$

Question Number : 196 Question Id : 6406531279912 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Find the value of $P(Z \leq 1 \mid Y = 3)$. Enter the answer correct to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.812 to 0.818

Question Id : 6406531279913 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (197 to 198)

Question Label : Comprehension

The joint PMF of two discrete random variables X and Y is given in the following table:

Y \ X	0	1	2	$f_Y(y)$
0	0.1	a	0.1	0.35
1	0.2	0.3	a	b
$f_X(x)$	0.3	0.45	c	1

Define a random variable: $Z = X^2 + Y - 1$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 197 Question Id : 6406531279914 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What are the values of *a*, *b*, *c* ?

Options :

6406534314857. ✔ *a* = 0.15, *b* = 0.65, *c* = 0.25

6406534314858. ✖ *a* = 0.25, *b* = 0.65, *c* = 0.25

6406534314859. ✖ *a* = 0.15, *b* = 0.75, *c* = 0.25

6406534314860. ✖ None of these

Question Number : 198 Question Id : 6406531279915 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following option(s) is/are correct?

Options :

6406534314861. ✔ Range of $Z = \{-1, 0, 1, 3, 4\}$

6406534314862. ✖ The value of $P(Z = 3 \mid Y = 0)$ is 0.25

6406534314863. ✖ Range of $Z = \{-1, 2, 4, 5\}$

6406534314864. ✔ The value of $P(Z = 4 \mid X = 2)$ is 0.15

Question Id : 6406531279916 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (199 to 200)

Question Label : Comprehension

A package delivery time *T* (in hours) is uniformly distributed over the interval [*a*, *b*]. It is known that the probability that the delivery takes less than 5 hours is equal to the probability that it takes more than 11 hours.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 199 Question Id : 6406531279917 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Find the value of *a* + *b*.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

16

Question Number : 200 Question Id : 6406531279918 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

If the value of b is 14, then find the value of $P(5 < T < 11 \mid T > 7)$. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.54 to 0.60

Question Id : 6406531279919 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (201 to 202)

Question Label : Comprehension

A freelance designer receives different payments from her clients based on the type of project. Based on the past data:

- She earns ₹18000 in 40% of the projects.
- She earns ₹10,000 in 35% of the projects.
- She earns ₹5,000 in 15% of the projects.
- She earns nothing in the remaining projects.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 201 Question Id : 6406531279920 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

What is the expected payment that the designer earns for a randomly chosen project?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11450

Question Number : 202 Question Id : 6406531279921 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Using Markov's Inequality, find an upper bound on the probability that the designer earns at least ₹15,000. Enter the answer correct up to three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.760 to 0.766

Question Id : 6406531279922 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (203 to 204)

Question Label : Comprehension

Let $X, Y \sim$ i.i.d. Uniform$\{1, 2\}$ be discrete random variables. Define $U = X + Y$ and $V = X$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 203 Question Id : 6406531279923 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Calculate the value of $\text{Var}(2X + 1)$.

Options :

6406534314869. ✖ 1.5

6406534314870. ✖ 0.5

6406534314871. ✔ 1

6406534314872. ✖ 0.33

Question Number : 204 Question Id : 6406531279924 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Find the correlation coefficient between *U* and *V* . Enter the answer correct upto two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.68 to 0.74

Statistical Computing

Section Id :	64065391720
Section Number :	12
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	10